



TÉCNICO SOLAR BOAT



Electrical system overview for SG01

Lisbon, September 2025

Afonso Daniel Guerreiro Coelho

Contents

Acronyms	3
1 Revision History	5
2 Introduction	6
2.1 Regulation restrictions	6
2.2 TSB's few standards	9
3 System architecture	12
4 Low Voltage (LV) systems	15
4.1 LV Battery box	15
4.1.1 Low Voltage BMS	16
4.2 AUX LV Battery box	17
4.3 Electronics Plate	18
4.3.1 Flight control	18
4.3.2 Telemetry	19
4.3.3 Data logging	20
4.3.4 Dashboard	21
4.3.5 Fuse/relay box	22
4.3.6 Raspberry Pi	23
4.3.7 CAN Breakout board	23
4.4 MPPT box and Solar panels	24
5 High Voltage (HV) systems	27
5.1 HV safety measures	27
5.2 HV Battery box	28
5.2.1 Battery Packs Assembly	30
5.2.2 BMS	31
5.2.3 Auxiliary PCB	31
5.3 Motor	32
5.4 Motor Controller	32
6 Future Work	35
7 Full Boat Overview	36
8 LV Battery Box Schematic	37
9 HV Battery Box Schematic	38
10 Electronics Schematic	39

Acronyms

AHRS	Attitude and Heading Reference System
AoA	Angle of Attack
BMS	Battery Management System
CAN	Controller Area Network
DBC	CAN database
GUI	Graphical User Interface
HV	High Voltage
HVD	High Voltage Disconnect
IMD	Insulation Monitoring Device
LV	Low Voltage
MEBC	Monaco Energy Boat Challenge
MPPT	Maximum Power Point Tracker
PCB	Printed Circuit Board
PID	Proportional–Integral–Derivative
SDC	Shut Down Circuit
SG01	São Gabriel 01
SM01	São Miguel 01
SM02	São Miguel 02
SP01	São Pedro 01
SR01	São Rafael 01
SR02	São Rafael 02
SR03	São Rafael 03
TSAL	Tractive System Active Light
TSB	Técnico Solar Boat
USB	Universal Serial Bus
VPS	Virtual Private Server

1 Revision History

SG01 - 21/09/2025 - Afonso Coelho

2 Introduction

This document aims to explain plainly how the electrical system architecture of the vessel São Gabriel 01 (SG01) for the 2024-2025 season is and how it all works. Additionally, there are two more documents explaining how every costume PCB works, how we manufacture our solar panels, and a follow-up to our open-source GitLab with all Printed Circuit Board (PCB) code ([Everything will be available to the public on Christmas Eve](#)).

2.1 Regulation restrictions

First, we will explain the main restriction imposed by the regulation of the Monaco Energy Boat Challenge (MEBC) for the Sealab class.

There are a few very important points:

- Point 6.1 Voltage - That auxiliary system voltage *is* $\leq 52V$. - The voltage for each module (battery, fuel cell, inverter, motor etc.) *is* $\leq 600V$.
- Point 6.2 Isolation - electrical connections between motors, batteries, inverter, and solar panels, wind turbines (if applicable) are isolated and safely connected.
 - There is real-time monitoring of isolation. A continuous green light should shine if isolation is above $V_{max} * k\Omega$ (In case of 400 VDC, minimum $0.4M\Omega$). A continuous red light should show, and an audible alarm should be heard if isolation is below this value. This light must still work if Emergency Stop Button (see 6.12) is engaged.
 - Isolation gloves are onboard.
 - Isolation hook is kept in the working area ashore, Allowing a person to safely remove an electrified person.
 - High voltage systems should be electrically isolated.
 - High voltage systems should be physically isolated in order to avoid accidents while working on the boat.
- Point 6.3 Fuses - There are fuses between battery $\leftrightarrow$ motor and solar panel, wind turbine $\leftrightarrow$ battery (if applicable).
- Point 6.4 Grounding - Grounding is accessible from outside.
- Point 6.5 High Voltage Handling Certification - A training was attended by all of those who will work on high voltage system. This certification can be issued by any training centers or by the university.
- Point 6.6 Cabling - The cables are isolated, and that cross-sectional areas respect Table 2 (see at the end of this document). If an equivalent national standard is used, this needs to be shared with the technical committee along with this checklist. In case cables are designed differently, datasheet must be provided.
- Point 6.8 Batteries - are situated $\geq 1m$ away from any person onboard.
 - are placed in a separate, non-conducting, IP 57 or higher housing.
 - are liquid cooled or forced-ventilated.
 - have ventilation inlets and outlets $\geq 1m$ away from any person onboard.
- Point 6.9 Non-commercially available batteries - If applicable, please notify technical committee at registration stage. Application may be refused at Organisation's discretion.

- Point 6.10 Battery management system (if applicable) - When using other than lead-acid or lead-gel batteries, a battery management system is installed which:
 - guards the battery pack voltage and temperature.
 - guards each individual cell voltage and current.
- Point 6.11 Solar panels - Solar panels are mechanically secure and have protection on sharp corners.
- Point 6.12 Emergency Stop Button - When activated, the switch should stop and isolate all electrical components from the boat (apart from the real-time control of the isolation – see 6.2).
- Point 6.13 Battery Service Disconnect - Activating the service disconnect button fully disconnects and isolates the main battery from the rest of the boat. The service disconnect button segments the battery into sections with voltages below 60 V to allow safe work on the battery pack. If this was not possible, these safety aspects should be demonstrated beforehand to the organizers.
- Point 7.2 Fire detector - One fire detector installed in each motor, battery, and fuel compartment.
- Point 7.3 Fire extinguisher - One manual multi-purpose fire extinguisher to a recognised standard in cockpit or within reach of helmsman.
- Means of communication between liaison and pilot
 - Works at any range with mobile coverage, without WiFi, or grid connection.
 - The means of communication must be integrated into the helmet of the pilot, be waterproof and supplied with a battery of sufficient capacity to last at least one race without being recharged.

Since the Sealab class is very recent, the regulations are not very demanding, but this was the first high-voltage vessel from Técnico Solar Boat (TSB), so we had to follow some industry standards and safety rules present in the Formula Student Rules. The chapters most relevant for the electrical systems are T11 and EV. Here are some of the most relevant rules:

- T11.1.3 All LVS parts must be adequately insulated
- T11.1.6 The LVS must not use orange wiring or conduit.
- T11.1.7 The LVS must be grounded to the chassis.
- T11.3.1 An LVMS according to T11.2 must completely disable - power to the LVS
- T11.4.2 Each shutdown button must be a push-pull or push-rotate mechanical emergency switch where pushing the button opens the SDC, see EV6.1 and CV4.1.
- T11.4.3The buttons must be easy reachable from outside the vehicle.
- T11.4.6 Shutdown buttons must be rigidly mounted to the vehicle and must not be removed during maintenance.
- T11.7.5 Completely closed LV battery cases must have an overpressure relief. Venting gases must be separated from the driver by a firewall.
- T11.7.6 LV batteries must be protected from short circuits, not more than 100 mm from ungrounded terminals

- EV3.1.3 An electrically conductive part is grounded if its resistance to LVS ground is below 100 m Ω , measured with a current of 1 A, and the grounding wire is able to continuously carry at least 10 % of the TS accumulator main fuse current rating.
- EV3.1.4 Parts of the vehicle which are or may become electrically conductive within 100 mm of any TS component must have a resistance below 100 Ω to LVS ground.
- EV4.3.1 The entire TS and LVS must be galvanically isolated, see EV1.2.1 and IN4.1.1.
- EV4.3.1 TS and LVS circuits must be physically segregated such that they are not running through the same conduit or connector, except for interlock circuit connections
- EV4.3.6 If TS and LVS are on the same PCB, they must be on separate well-defined areas of the board, meeting the spacing requirements in table 5, each area clearly marked with “TS” or “LV”. The outline of the area required for spacing must be marked. Grooves and cut-outs must have a minimum width of 1.5 mm to influence the creepage path. “Conformal coating” refers to a coating insulator on a PCB. Solder resist is not a coating.
- EV4.5.2 Insulation material that is rated for the maximum TS voltage must be used. Using only insulating tape or rubber-like paint for insulation is prohibited.
- EV4.5.10 Every TS connector outside TS enclosures must include a pilot contact/interlock line which is part of the SDC. TS enclosures only used to avoid interlocks are prohibited.
- EV4.8.1 It must be possible to disconnect at least one pole of the TS accumulator by quickly removing an unobstructed and directly accessible element, fuse, or connector. It must be possible to remove the HVD without removing any bodywork. The HVD must be above 350 mm from the ground. Remote actuation of the HVD through a long handle, rope, or wire is not allowed.
- EV4.9.1 If a discharge circuit is required to meet EV6.1.5, it must be designed to handle the maximum TS voltage permanently. After three subsequent discharges within 15 s in total, the discharge time specified in EV6.1.5 may be exceeded. Full discharging functionality must be given after a reasonable time with a deactivated discharge circuit.
- EV4.10.1 The vehicles must include a single TSAL that must indicate the TS status. The TSAL must not perform any other functions. A TSAL with multiple LEDs in one housing is allowed.
- EV4.11.4 Closing the SDC by any part defined in EV6.1.2 must not (re-)activate the TS. Additional action must be required.

There are many important rules that have not been included here in order to keep this document from becoming too long. In any case, I recommend reading the Formula Student regulations, as I have not written down all the rules that may be extremely relevant to the design of a vehicle’s electrical system.

2.2 TSB's few standards

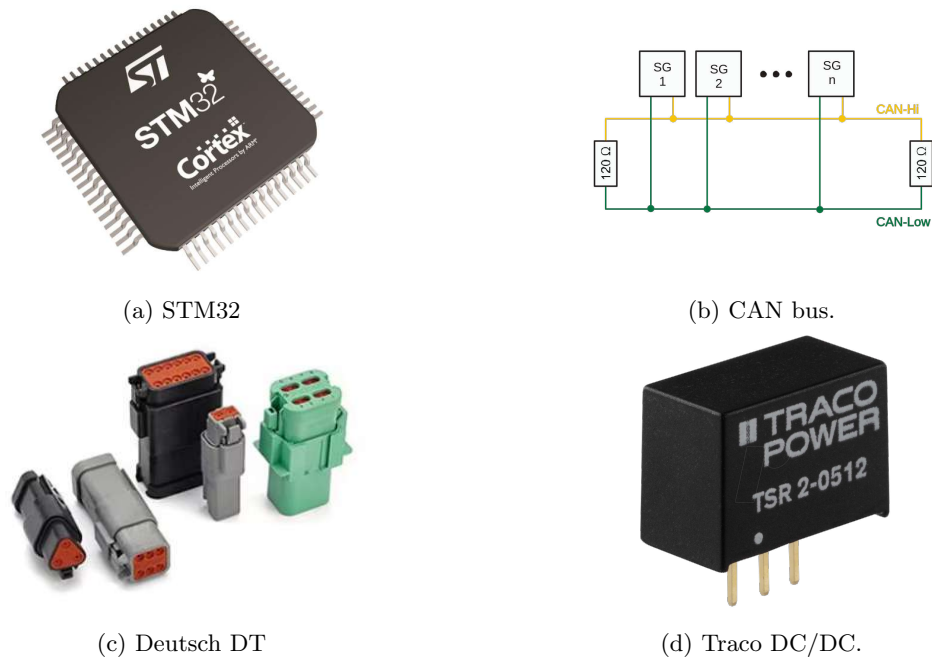


Figure 1: Some of our standards.

There are a few things that are standard at this time for the electrical system of our boats. Here a list of a few of them:

- **STM32** - For this boat, we are trying to convert every microelectronic to an STM32. But there are still a lot of PCBs with our old standard. Our older standard was Teensy 3.2, when a powerful chip was needed, or an AT90CAN otherwise. Since the chip of Teensy 3.2 was discontinued, we decided to change to STM32 because of its variety of chips with different characteristics.
- **Controller Area Network (CAN)** - This communication protocol allows us to connect any microcontroller to the bus and just decide whether to read its messages or not for each node individually. The first system we built used Universal Serial Bus (USB) and it was a mess of USB cables all going to a Raspberry Pi. We now have the CAN bus under control and it's very robust. For each microcontroller connected to the CAN bus we always use either the SN65HVD230DR transceiver for 3.3 V or the SN65HVD1050DR for 5 V logic. All AT90CAN should use 5 V transceivers (for simplicity).
- **CAN connections** - PCB connections are all made with Micro-Fit+ connectors (from Molex), each PCB should have only one connector and a CAN Breakout should be present in each box. For connections between boxes, a HDSCS connector with tow positions (from TE connectivity) is used. The old standard was an RJ45 connector that was not practical to assemble but very robust (there are still some old PCBs with this connector).
- **Connectors** - for signal between boxes, the relatively cheap connectors from TE (Series Deutch DT) can be received as samples or bought. For the connectors exposed to the elements, LEMO (Series K) is used because they are more robust and salt water resistant, but very expensive (a sponsor is normally possible, but must be ordered early in the year, preferably around October). Some more specific connectors are

also used, such as the BPC series from Phoenix Contact (for power from the battery box to the Aux box). The tractive system is composed by an AMP+ HVP800 to connect the battery to the motor controller. Than some SurLok Plus™ Series connectors are used to connect the motor controller to the motor. For solar panels, we use MC4 connectors from TE Connectivity For LV charging, we use the POWERPACK LV1000 from Aptive, and the same is used to connect MPPTs box and AUX box. For HV charging, the AMP+ HVA 1200 is used.

- PCB connectors - We use Molex connectors, since we are sponsored by Molex. This year, with new challenges, we adopt new standards, and so the Micro-fit+ series was chosen for its robustness, variety of inputs, and ease of assembly. There are still some PCBs using the old standard Micro TPA and Mini-lock
- 24 V power for PCBs - all PCBs are powered by the 24 V bus and have their own 5 V or 3.3 V converter from Traco Power TSR series DC/DC's.
- Connector Polarity - Pin 1 of every connector that carries power (regardless of the voltage) should **always be the positive!** Check connector data-sheet to be sure which pin is the number one, pin 2 should be GND. If the connector has 4 pins and will be only used for power the lowest pins (1 and 2) should be the positive and the highest pins (3 and 4) should be GND. This standard is valid whether we are talking about PCB or box connectors.
- CAN Connector Polarity - Pin1 is always CAN Low and pin 2 CAN High.
- Fusing - Fuses should be placed in each PCB on the low voltage side, after the 24 V DC/DC such fuses can help to save a lot of electronics in case the wrong connections are made. The rating of such fuse should be calculated based on the current consumed by each PCB low voltage bus.



(a) Deutsch DT



(b) LEMO's Series K.



(c) BPC series



(d) TE's AMP+ HVP800.



(e) TE's AMP+ HVA1200.



(f) MC4



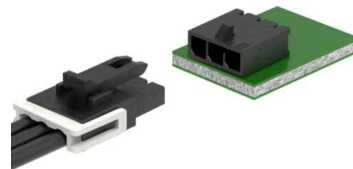
(g) Powerpack LV1000



(h) SurLok Plus.



(i) Molex's MicroTPA.



(j) Micro-fit+

Figure 2: Most of the connectors used on SG01.

3 System architecture

The system is separated in 6 main boxes, each one of these has its own electronic parts which will all be explained in detail.

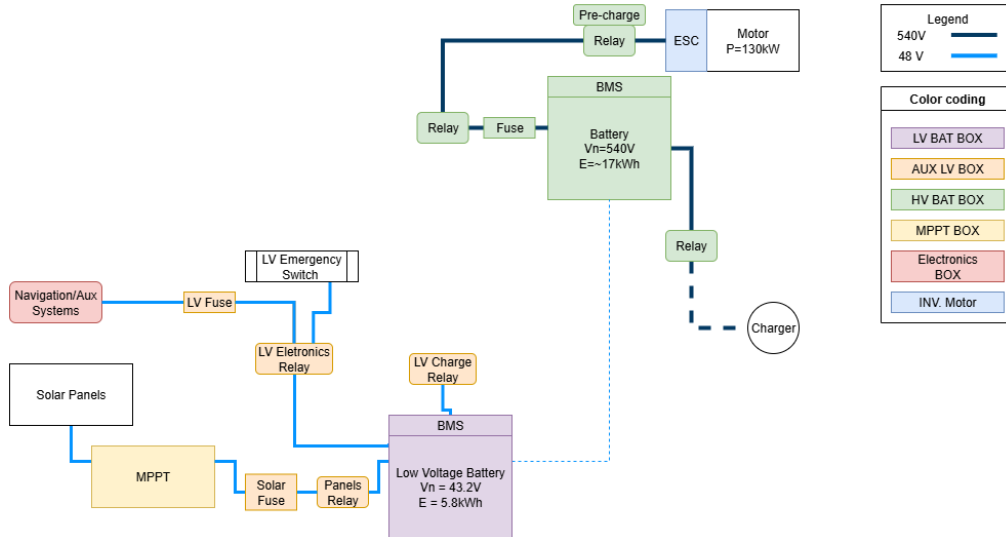


Figure 3: Simplified electrical schematic of SG01

The system is divided in two main blocks, LV systems and HV systems. The two blocks are interconnected but galvanically isolated, meaning that the 0V from the HV system is not ground. Besides power interconnection, there is a CAN bus connecting all the boxes, so they all communicate and can listen to whatever any microcontroller sends to the CAN bus.

The LV system is responsible for all the auxiliary systems like dashboard, telemetry, flight control, etc. Some of those are essential for a clean and secure navigation, so this system always needs to be ready to work.

There are 4 boxes and some scattered components that together make a LV system:

- LV Battery: responsible for storing and give energy to the system;
- Aux LV Box: this box contains the electronics, charge and Solar panels relays which, with the help of the Battery Management System (BMS), will protect the battery. Additionally, it has two DCDCs from Traco power to provide 24V and 12V to the system;
- Electronics Plate: responsible for powering all of the boat's components, flight control, data logger, collecting all the information acquired by the boat's sensors and boards, and sending it to the pilot and our telemetry system;
- MPPT BOX: responsible for acquiring all the energy generated by the solar panels;
- Out of box components: Dashboard, throttle, Attitude and Heading Reference System (AHRS), ultra-sounds, inverters, lights, horn, emergency switch, antenna, Bilge pumps and solar panels;

The standard voltage for the LV power is 24V because of its steady voltage given by a DCDC. But when the components support the all range of voltages of the LV battery, connecting directly to the battery is preferred. For now, only the nemas and the inverters that power the cooling system are operating at battery voltage. As for the 12V, we will always need a small DC-DC to power some electronics but can be reduced in the future.

The components operating at 12V are: Bilge pumps, Radio, ultrasounds, PCB of ultrasounds, and the LV side of the motor controller. Additionally, the bilge pumps and some LEDs are connected to 12V too (this was only done because of the lack of stock of some suppliers).

As for the HV system, special care is needed because of its risks. Some safety measures were taken, mainly based on Formula Students regulations. All of these measures will be explained in detail further in this document.

This system is responsible for the tractive system, in other words, to make the boat move. This system is mainly composed by:

- HV Battery Box: Stores energy and powers the motor. It also contains all safety systems associated with the battery;
- Motor Controller: Converts the constant voltage given by the battery to a 3-phase signal, which will control the torque, current and RPMs of the motor;
- Motor: Converts the energy given by the Motor Controller to kinetic energy;
- Out of the boxes safety equipment: kill switch, TSAL lights, IMD lights and grounding.



Figure 4: Dashboard

The pilot has 4 switches at his disposal to turn on/off systems on the boat. Two LV switches, the Emergency and the electronics switch. The Emergency switch, as the name implies, should only be used in emergencies, while the mains switch is the switch used to shut down the entire boat. This is due to the limited cycle life of the Emergency button. Both of these switches turn off the HV by shutting down the LV system (the HV system automatically turns off if the LV system is shut down). There are 2 switches to control the HV system too, the kill switch and the HV service switch. Both of these switches disconnect the connection between the battery and the motor controller, keeping the HV only in the battery box. There are two switches for the same reason as the LV system and to add an extra layer of safety. In this way, there needs to be 2 actions to turn on the HV system.

We should place two more emergency SW in the boat. This boat is too big to always have access to only one switch

Additionally, there are 5 switches to control the state of the bilge pumps (manual, off, or automatic).

The rest of the systems are controlled by the driver in the dashboard. There is more information about the dashboard further in the document

The navigation lights should be activated in the dashboard. Quando isso funcionar mete se aqui um textinho sobre isso também

4 LV systems

4.1 LV Battery box

The low voltage battery box only contains the battery and is own costum BMS. This battery was made for São Miguel 02 (SM02) and repurposed for SG01.

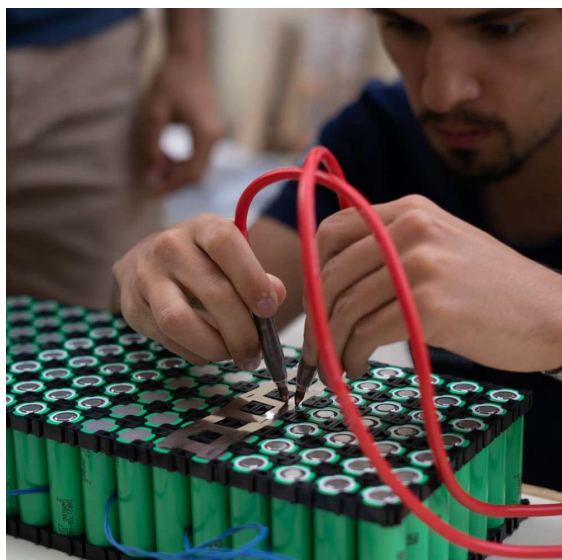
This battery uses Samsung INR21700-50G and have a 12s28p configuration. Its max voltage is 50.4V which is lower than 52V (maximum imposed voltage by the competition)

Max Voltage	50.4	V
Nominal Voltage	43.2	V
Min Voltage	30	V
Max Power	11733.12	W
Capacity	5855.56	Wh
Number of cells	336	
Weight	25200	g
Commercial Price	991.2	EUR

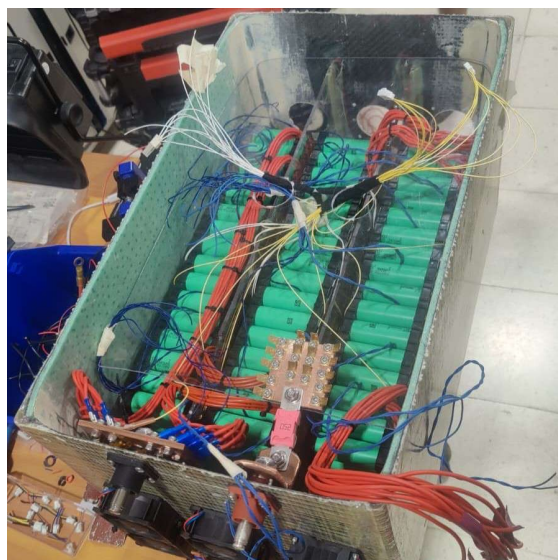
Table 1: LV Battery specs (only considering cells)

Using Samsung INR21700-50G 3 packs were assembled by the team, each with a 4s28p configuration. Each pack has thermistors monitoring cells temperature and monitoring the temperature of the nickel tabs joining cells.

The nickel tabs were soldered to the battery cell poles using our spot welder (the kWeld + kCap).



(a) Spot welding



(b) Battery final assembly *Meter uma foto decente*

Figure 5

The packs are then interconnected with cables and finally connected to a custom copper piece that connects the battery to his connectors. The previous connectors used were the Radloks by Amphenol, but this year we decided to change them to SurLok Plus because we were already using this connectors in HV battery. The Radlok series is verry similar to the SurLok Plus series, we only changed it because we were able to get free samples from Phoenix Contact to save money.

4.1.1 Low Voltage BMS

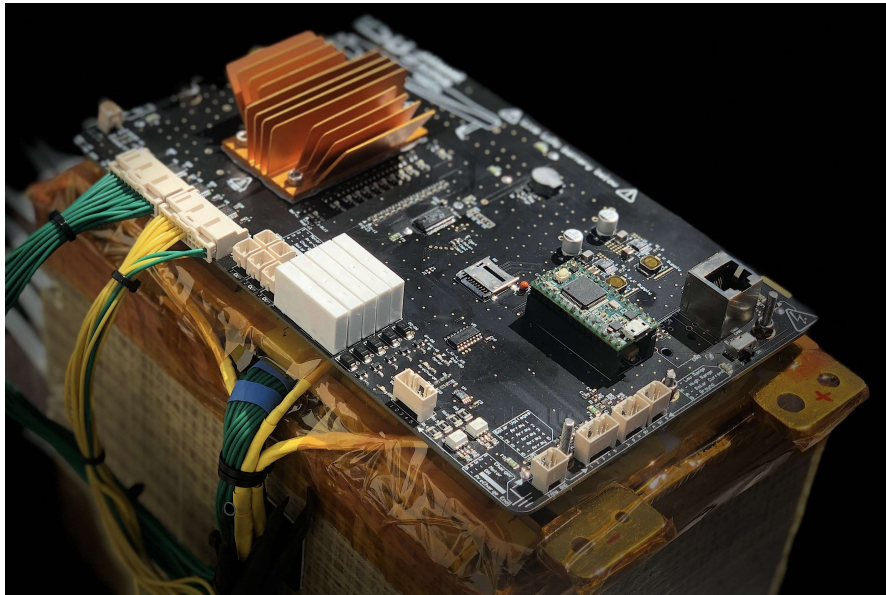


Figure 6: SR-03's BMS assembled on the top of the battery. [Mudar a foto para uma do SG01](#)

The BMS is responsible for making sure all the battery parameters are within the safe limits, and makes some extra management of relays for the solar panels and charging. Keep in mind that this battery only powers the auxiliary systems.

It measures all cell voltages, about 40 temperature points, battery current (with the help of the CAB500), total solar panels current, senses if there is voltage in some points, for example to know if a charger has been connected.

The BMS controls all relays on the low-side, for example, if the voltage gets too low it will open the load relay (in this battery this is not applied because the load is the auxiliary system), if the voltage gets too high it will open the solar panels and charger relays. It also sends all the measurements it makes to the CAN bus.

The BMS present on SG01, was already used in previous boats like SM02, São Rafael 03 (SR03) and São Pedro 01 (SP01). Originally made by Sebastião Beirão in his master's thesis for SR03.

Besides the hardware we also have software to see the data read by the BMS. Figure 7 shows a screenshot of the graphical interface made using QT5. This app allows us to see all cell voltages, temperatures, relay states, currents, solar panels data, program some BMS parameters and more by simply connecting the Teensy's USB port to a computer.



Figure 7: Graphical User Interface (GUI) for the BMS data.

A more thorough description of the BMS can be found in Sebastião's Master Thesis "Battery Management System for Solar Powered Vehicles applied to Técnico Solar Boat prototype" available in [Técnico's web page](#).

4.2 AUX LV Battery box

As the LV battery was done for SM02, some auxiliar components were missing in the battery box, like the DCDCs and relays. To make the most of what had already been done, an auxiliary box was created with the solar, charger and electronic relays, DCDCs to convert 48V into 12 and 24V and two current sensors to measure the power supplied by the panels and the power from the DCDC that charges the LV battery from the HV battery.

The DCDCs used are TEQ 300-4815WIR and TEQ 300-4812WIR from Traco Power (our sponsor). Both DCDCs have a potentiometer to adjust the output precisely. If the 24V or 12V power bus are showing an incorrect value, this should be the first approach to the problem. Other approach could be using the sense pins, by placing them near the load.

Using Te connectivity relays, specifically the IHV40A5ANG (24V coil) and IHV40A6ANG (48V coil) models, the BMS can control the power supply to the system. For example, if the battery is overcharged, it opens the relay for the solar panels and the charging panel, and if the battery is overloaded or under voltage, it opens the electronics relays (in this case, this is not done due to the load being the auxiliary systems). The same reasoning applies to overtemperature and overload.

Two CAB500 from LEM (our sponsor) are used to measure the current of the solar panels and the overall current. These current sensors are very popular in the automotive sector due to their robustness and CAN communications.

4.3 Electronics Plate

4.3.1 Flight control

Regarding hardware

The flight control system has direct access to the state of the boat, meaning all values used by the controller are obtained from the boat's sensors. The AHRS is used to measure Euler angles, angular velocities, and linear velocities. Additionally, a combination of ultrasonic sensors is used to estimate the boat's height relative to the water. These ultrasonic sensors have a dedicated PCB that reads the analog values and feeds them into the CAN bus.



(a) UGT519, IFM ultrasonic sensor

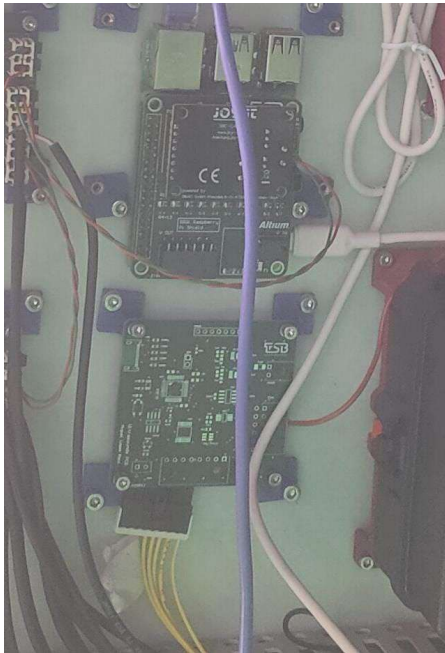


(b) MTi-680G-SK, form Movella

Figure 8: Sensors used for flight control.

The system uses a wing design to generate lift. Instead of three hydrofoils, it employs a front wing equipped with two ailerons and a rear hydrofoil. These serve as the actuators of the system, with NEMA stepper motors adjusting their angle of attack through the CAN bus.

All sensors and actuators values are read and controlled by a Raspberry Pi. For flight control, a C++ executable runs on the Raspberry Pi, reading CAN bus values and setting the appropriate angles of attack.



(a) PCBs used in the flight control meter
uma foto parecida de jeito



(b) Nanotec and maxon Nemas for actuation

Figure 9

Regarding Algorithm

The C++ executable runs the control loop at a constant sampling rate. It supports multiple control modes, each with a dedicated controller and independent gains. The main mode is the full-control mode (while at flight), where a is applied, using the full state of the system to control all actuators. This is possible because the boat's dynamic model was formulated by our team.

Other modes decouple the system. Due to the design, roll can be controlled through the aileron angles, while height can be roughly controlled with the rear foil. In these cases, a Proportional–Integral–Derivative (PID) controller is applied to each subsystem.

Additionally, specific controllers are implemented for *takeoff* (lifting the boat from full contact with the water) and *shutdown* (transitioning from flight mode back to full contact with the water). These modes are necessary due to the size of the vessel and to ensure a smoother ride for passengers. They also rely on PID controllers, but with a ramp input for height, providing a smoother start and end to operation.

Although these multiple modes increase the complexity of both the code and the control algorithms, they provide flexibility to switch between controllers for testing. The design objective is to maximize the modularity and testability of the control system.

4.3.2 Telemetry

The telemetry system is designed to give the ground team a detailed, real-time view of everything happening inside the boat, allowing them to monitor performance, analyze data, and alert the pilot if any abnormal behavior is detected. Process Overview:

1. **Onboard Data Collection:** A Raspberry Pi installed in the boat collects data from all components via CAN bus;

2. **Data Transmission:** The Raspberry Pi sends this information through HTTP POST requests to our server infrastructure;
3. **Server & Storage:** A Virtual Private Server (VPS) hosts an HTTP listener, which receives the incoming data and writes it into InfluxDB.
4. **Database Management:** InfluxDB is used to efficiently store and organize the time-series data from the boat.
5. **Visualization:** The data in InfluxDB is connected to Grafana (running on the VPS), which provides the team with a detailed and customizable dashboard for analysis and monitoring.

This setup ensures that the ground team has continuous access to precise operational data. By combining real-time monitoring with historical data visualization, the telemetry system plays a key role in ensuring safety, reliability, and performance optimization of the boat.

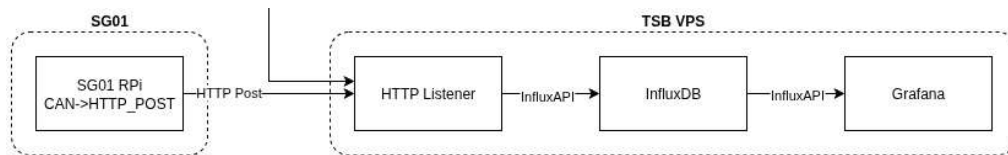


Figure 10: Telemetry system architecture from boat to VPS

4.3.3 Data logging

The Kvaser is a device that we have a lot of experience with and it has proven to be really important to our system. This device is able to read and save all the CAN messages on the bus so that they can be analysed later. This way, we can reduce the load on the micro-controllers, that in the past were saving this data to onboard micro-SD cards, which is a slow operation for the micros. Moreover, by using the Kvaser we ensure that acquired data is time synced. The device that we use, a **Kvaser Memorator Pro 2xHS v2** (Figure 11) supports simultaneous reading of up to two different CAN buses, which was really useful in previous boats like SR03 that had 2 can buses. In this boat, we are trying to concatenate the two buses into one, so this feature will not be needed. Kvaser can also be used as a CAN interface by using Kvaser's CANKing program and a CAN database (DBC)¹ file for the messages being read (DBC is not mandatory, it just makes your life easier). It is worth mentioning that it has an internal battery for time keeping that must be periodically charged by leaving its USB port connected to a wall charger overnight.

After the tests, the device should be connected to a computer and the log files saved and latter exported to .mat file so that they can be analysed. To export the .KMF files Kvaser's Memorator Configuration Tool and the respective DBC should be used. To analyse the data our Matlab Data Analysis App or the newly created Python based App should can be used.

¹A DBC is a file that has the translation between CAN messages and their physical meaning.



Figure 11: Kvaser Memorator Pro 2xHS v2

4.3.4 Dashboard

The boat is equipped with a digital dashboard designed to provide the pilot with a complete, real-time overview of everything happening on board.

The dashboard is displayed on a tablet screen and runs as a React application hosted on a Raspberry Pi.

The tablet was chosen because we couldn't find a good screen for the boat at an affordable price. And this tablet was free for us.

The Raspberry Pi is connected to a router to ensure constant internet connectivity. All boat component data is transmitted via CAN bus, and processed through our custom CAN library, which ensures the correct values are displayed to the pilot.

The dashboard is structured into a main page and six specialized subpages, easily accessible through a right-hand sidebar on the screen (show in Figure 12).

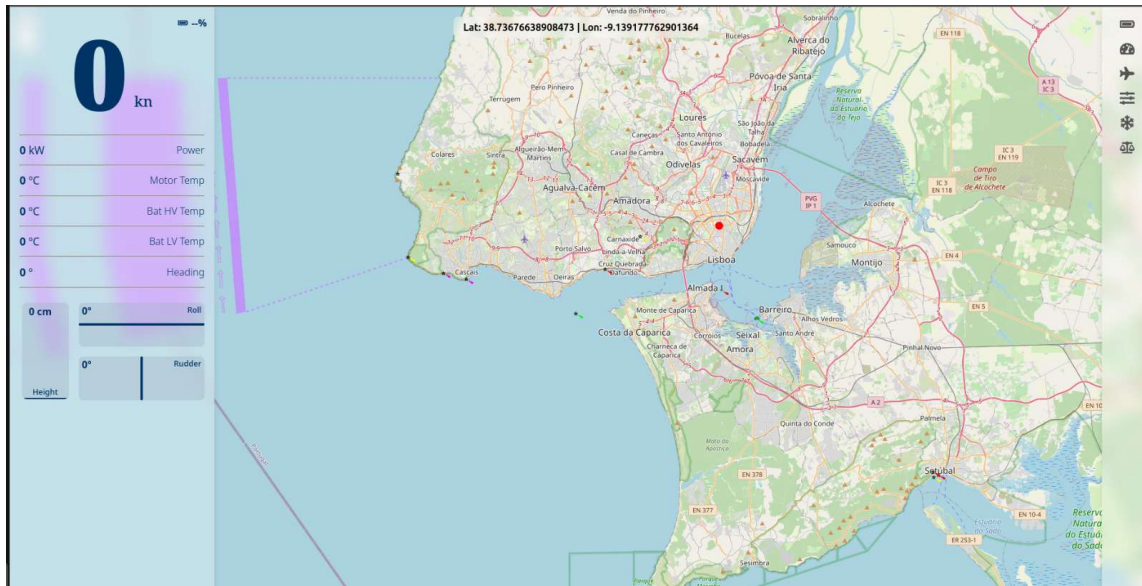


Figure 12: Dashboard Main Page

Main Page: Displays the most critical real-time information for navigation and operation. Key data includes velocity, power output, system temperatures, and a live map showing the current position of the boat.

Subpages: Provide detailed insights into specific systems:

1. **Battery Overview:** status, voltage and temperatures;
2. **Motor:** motor power, temperature;
3. **Flying Control:** monitoring stability and control systems,
4. **Throttle:** real-time input and response tracking,
5. **Cooling:** system temperatures and cooling efficiency;
6. **Cell Balancing:** ensuring battery health and equal charge distribution across cells;

This setup ensures the pilot can access both high-level and in-depth system data, enabling safe and efficient operation of the boat at all times.

4.3.5 Fuse/relay box

Fuses are one of the simplest and one of best protections that every circuit must have. In the past, a lot of PCBs were saved by fuses, and now we have a lot of fuses in the circuit.

SR03 had a dedicated PCB for fuses that served to distribute the power to every electronic and at the same time, taking some measurements of what was being used. For SG01 we wanted a more versatile way since we expected the system to grow in terms of number of sensors and PCBs, so we had the idea of using a Fuse box, similarly to cars and motos. In Figure 13 we see the fuse distribution and the actual fuse box that we used. This Fuse box is made by TE Connectivity and has a cover to make it IP67.



(a) Fuse and relay box

Color coding	PCB US	ETH SW	Bilge Pump 1	Ecrã	Nema 1	
48V	Motor controler	Rspb	Bilge Pump 2	Throttle	Nema 2	
24V	Radio	AHRS	Bilge Pump 3	Buzina	Nema 3	
12V	US	Kvaser	Bilge Pump 4		Nema 4	
	Router		Bilge Pump 5		Nema 5	

(b) Fuse box distribution **needs double check**

Figure 13

Similarly with the fuse box, we have two relays boxes that are controlled by the Raspberry Pi. The Raspberry Pi receives CAN messages or controls by the dashboard and turns on/off some on-board transistors, which in turn turn on/off the Relays. These relays are used for nothing in this moment, but we plan to use them for light control and/or control the cooling system.

4.3.6 Raspberry Pi

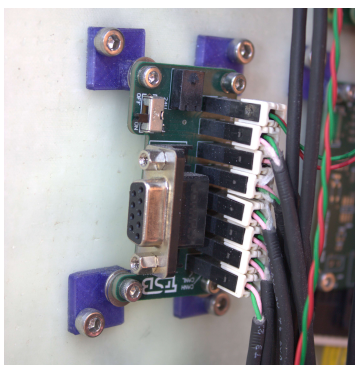
The Raspberry Pi 4 used in SG01 plays a critical role, as described in previous sections. It serves as the primary controller, executing the flight control code and acting as the microcontroller that communicates with both actuators and sensors via the CAN bus. In addition, it is responsible for telemetry by running a process that transmits selected CAN bus variables using MQTT packets. The Raspberry Pi also hosts a web-based dashboard and, finally, controls several relays through MOSFET drivers.

To enable reliable CAN communication and to interface with the MOSFETs used for relay control, a custom PCB was designed in the form of a shield for the Raspberry Pi. Further details regarding this PCB can be found in the dedicated PCB documentation.

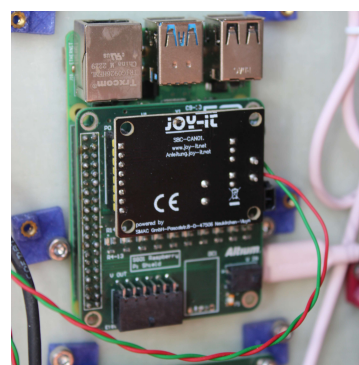
4.3.7 CAN Breakout board

Previously, we used a daisy-chain configuration for the CAN Bus inside the boxes, but this is not ideal since a loss of connection in the path would disconnect all the subsequent systems. For this reason, in SR03 a topology was chosen "inside the boxes".

This topology was kept in the SG01, but our standard can connector changed to a Micro-Fit+ with 2 circuits and a vertical header. So a new CAN breakout were made with 8 Micro-Fit+ connectors plus a DB9 Connector to connect a kvaser. Since it has a small battery, a power connector is needed to power kvaser.



(a) SG01 CAN breakout



(b) Raspberry Pi 4 shield

Figure 14

4.4 MPPT box and Solar panels

The solar panel configuration on the SG01 was designed to ensure maximum energy efficiency by optimizing solar exposure.

The total setup consists of a set of 120 Maxeon Gen 3 cells, model Me3, each with an approximate area of 155 cm². Regarding the panels, SG01 includes a total of 10 solar panels, distributed across 4 Maximum Power Point Tracker (MPPT)s from Genasun. The distribution was arranged to group into specific MPPTs, as detailed in Figure 15. where the yellow numbers represent the individual panels, and the red numbers indicate the corresponding MPPT. The connections are made through MC4 connectors, ensuring reliable and secure electrical connections.

The final distribution of the cells is outlined below:

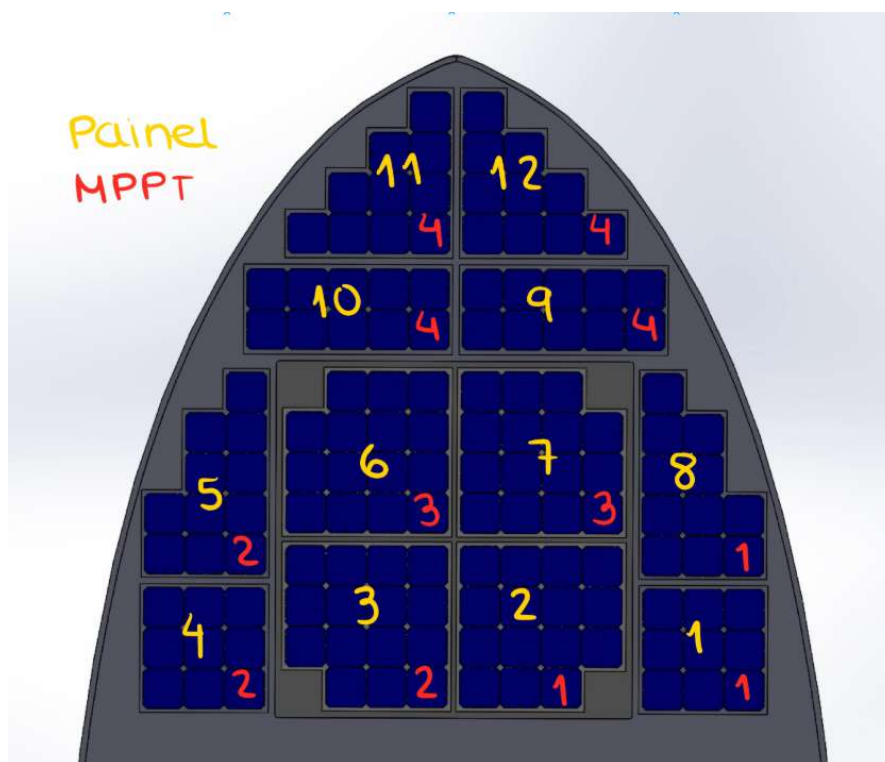


Figure 15: Distribution of Solar Panels and MPPT on SG01

The distribution of MPPTs considers areas of partial shading, temperature difference (mainly due to waves), and the relative power output of each solar panel. This arrangement not only minimizes the impact of varying solar incidence on the system but also allows for efficient management of the power generated by each panel.

Given that all the cells have the same area (0, 015506m²), the total area of the solar panels is $120 \times 0, 015506 = 1,8607\text{m}^2$. With these specifications, the photovoltaic system has the capacity to reach a maximum power of approximately 523.5 W under ideal conditions, considering an irradiance of 1000 W/m².

The selection of materials was made to ensure the durability of the solar panels. For SG01, the frontsheet selected is the CLR HDPY L, from Coveme. The panels will include a white backsheet, MONO L PLUS D, also from Coveme, which enhances efficiency by reflecting any light that passes through the photovoltaic cells, taking advantage of the bifacial properties of the cells. To encapsulate the cells and bond the frontsheet and backsheet, the Yparex Film 0F0035 encapsulant was chosen.

The distribution of the solar panels and MPPTs on SG01 was carefully planned to maximize efficiency and

meet regulatory requirements. The careful selection of materials and the robust lamination process made by the team, ensure optimized and reliable performance under real operating conditions.

If the lamination process has sparked your interest, you can also read our document on how we make our solar panels.

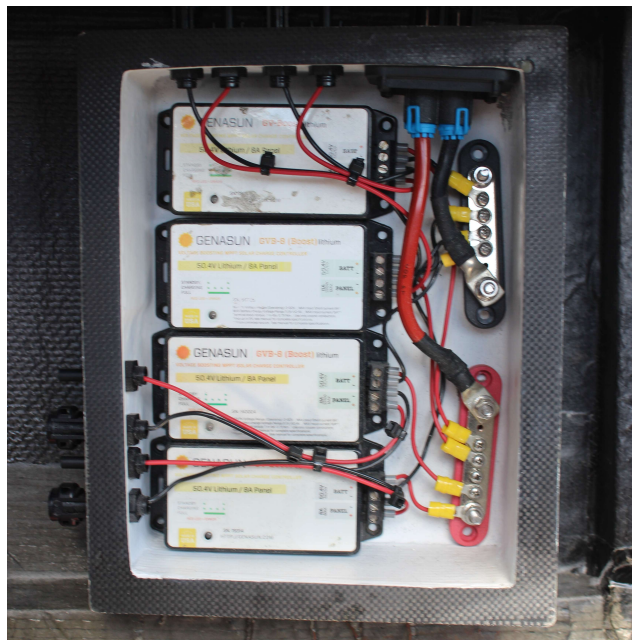
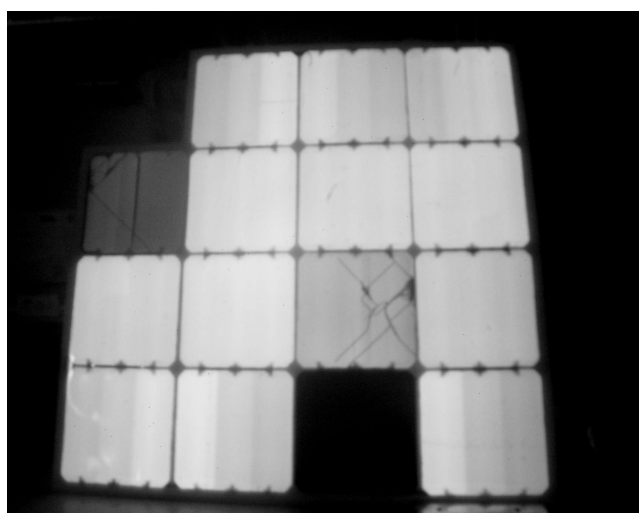


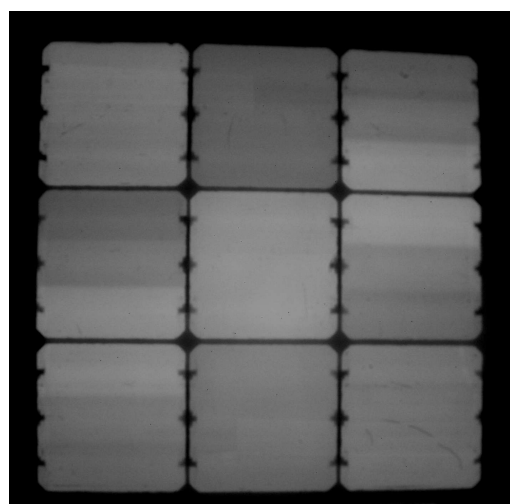
Figure 16: MPPT box

All the solar arrays are connected to our MPPT box (Fig. 16), where all the power produced by the solar panels is converted to a stable voltage to charge our low voltage battery.

Before installing the solar panels, electroluminescence tests were conducted. For the first time, these tests were performed in-house with the support of a former team member (Rui Oliveira). In previous years, the same procedure had been carried out at Open Renewables using professional equipment.



(a) Worst results



(b) Best results

Figure 17: Electroluminescence test results

These tests consist of applying a controlled electrical current to the solar cells and capturing the emitted infrared radiation with a specialized camera. The resulting images make it possible to identify micro cracks, broken cells, and other structural or manufacturing defects that are not visible to the naked eye, thereby ensuring the quality and reliability of the panels before integration into the system.

”To capture the images, a Nikon D70 camera will be used, equipped with a Nikon DX Format 6.1 effective megapixel CCD sensor, with its IR filter is removed. This modification enables the sensor to capture radiation in the NIR region. Although the quantum efficiency in this range is relatively low, it is sufficient to produce high-quality images to detect defects and aging processes in solar cells.” ”The Nikon D70 configuration was first adapted from similar setups in the literature and then fine-tuned through trial and error to improve image quality with this setup. The final settings were an exposure time of 30 seconds, an aperture of f/5, ISO 1000, and a shutter speed of 1/2.” Quotes taken from Rui Oliveira’s thesis. When the thesis is published, we will post a link here.

The setup used in Rui master’s thesis is summarized in Figure 18. In our case we did no heat the panel with a Peltier module nor did we measure the temperature.

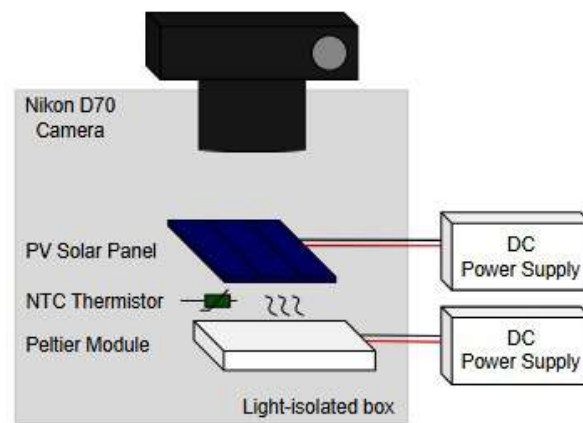


Figure 18: Electroluminescence tests setup

5 HV systems

5.1 HV safety measures

This year, the main innovation for our team was the high-voltage system. As the voltage increased from previous batteries (24 or 48V), the danger increased too. Having that in account, the team studies how to keep it as safe as possible.

These measures begin with safety equipment (gloves class 00, isolation mat, glasses, isolation hook, etc.). The system can always fail, and this safety equipment will protect the worker from the electrified equipment.

Then there are the always active zones and the cushion zones. The cushion zones are all the zones where cables and connectors are outside boxes. These are areas that require considerable care in their construction and maintenance, but there should be no high voltage on the outside. In SG01 there is only one always active zone, inside the battery. When assembling the battery or working near it, it is mandatory to use the safety equipment and always be aware of your surroundings. In the next section, there is more information about the battery and its safety.

In order for any component outside the battery box to be powered by high voltage, several safety measures must be in place. In practice, this is done with a Shut Down Circuit (SDC) (Figure 19).

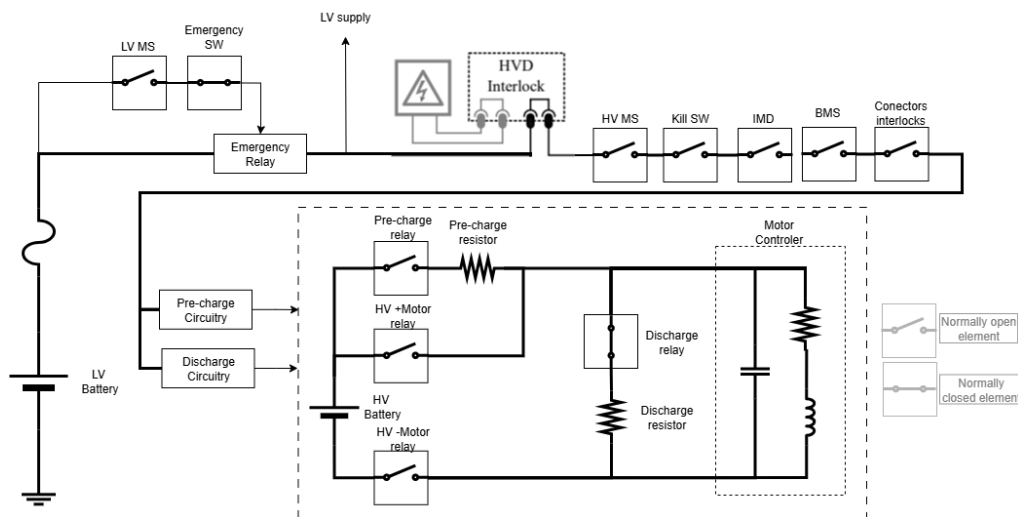


Figure 19: SG01 Shut Down circuit

The SDC is a 24 V signal that, at the end of the circuit, powers the pre-charge and discharge circuits, allowing the battery to supply power to components located outside the battery box.

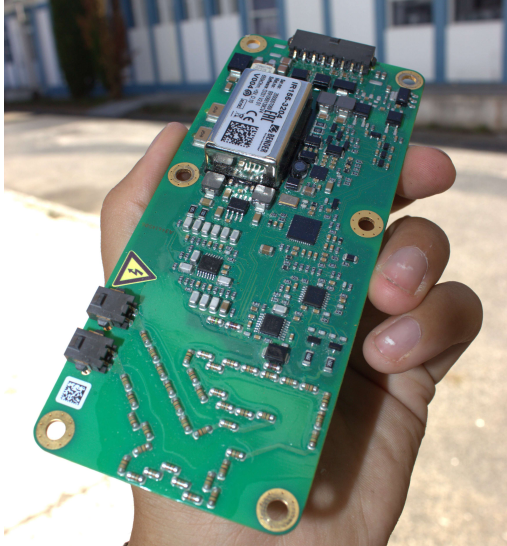
First, the signal came from the LV system that is only turned on if the LV Main Switch and the Emergency Switch are closed. Then the High Voltage Disconnect (HVD) (Figure 20b), this component provides a simple and secure way to disconnect the battery and all of the components inside and outside the box. The HVD is a switch+relay fixed to the battery box that mechanically disconnects the battery and its electronics. Besides that, the HVD has an Interlock that is present in the SDC in case of the relay contactor welds.

Then there is the HV Main Switch and the kill switch to disconnect the SDC manually or by a man overboard hazard.

The Insulation Monitoring Device (IMD) (Figure 20a) is the device that will only let the system work if there is no isolation failure. We use the IR155-3204 from Bender (sponsored by SISACOL) that is configured to detect failures when the resistance between HV+ and HV- is less than $650K\Omega$.

Even if all these measures are valid, it is still necessary for the BMS to detect that the batteries are in good condition to provide energy and that all connectors are properly connected. This is possible through the interlocks present in all HV connectors.

Note: the components in the SDC can be out of order since they are all in series.



(a) IMD



(b) HVD

Figure 20

5.2 HV Battery box

The battery box is the most dangerous zone of all the boat. One of the main safety requirements for this battery was to divide the battery in 10 low voltage packs, each with a 15s7p configuration, resulting in a 150s7p battery. With this modular architecture is possible to assemble each pack safely and do its maintenance without the presence of HV, by just disconnecting the packs.

The battery system was not designed with the objective of maximizing energy storage capacity, as the primary energy source of the project is intended to be a hydrogen-based power supply (not yet implemented). Anticipating the integration of a hydrogen fuel cell, the battery pack was developed with a primary focus on achieving a high charge acceptance rate. This characteristic is essential, given that hydrogen fuel cells exhibit limited responsiveness in rapidly reducing power output. Consequently, during transient conditions where energy production exceeds demand (like when stopping), the surplus energy must be absorbed by the battery rather than directed to the motor controller, while the fuel cell output gradually decreases.

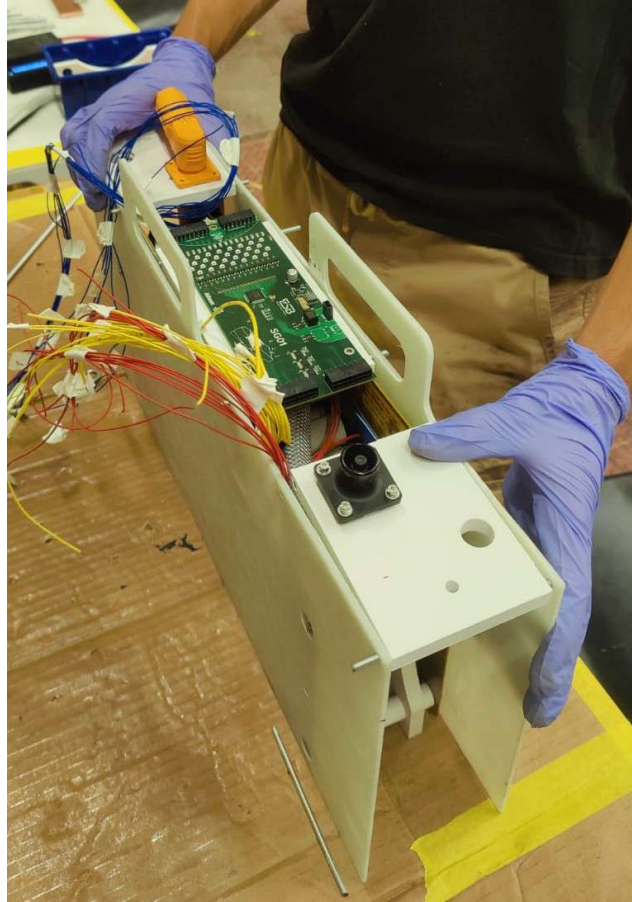


Figure 21: Battery Pack substituir por uma foto da bat aberta

Based on these requirements, Molicel INR21700-P45B lithium-ion cells were selected. Each cell provides a nominal capacity of 4500 mAh and supports a continuous charge rate of 9 A, making them suitable for high charging rate applications. The cells are cylindrical in format, which offers enhanced mechanical robustness and safety compared to pouch-type lithium-ion cells. Furthermore, prior experience with cylindrical cell soldering facilitated their integration into the current design.

The final battery pack achieves a nominal voltage of $3.6 \times 150 = 540V$. Its total nominal energy capacity is calculated as $4.5 \times 3.6 \times 150 \times 7 = 17010Wh$. The selected cells support a discharge rate of 10C, which corresponds to a maximum continuous discharge power of $4.5 \times 10 \times 3.6 \times 150 \times 7 = 170100W$. In terms of charging capability, the pack supports a maximum charge power of $9 \times 3.6 \times 150 \times 7 = 51030W$. The summary of the results is represented in table 2.

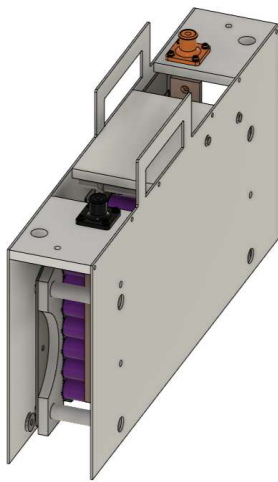
Max Voltage	630	V
Nominal Voltage	540	V
Min Voltage	375	V
Max Discharge Power	170100	W
Max Charge Power	51030	W
Capacity	17010	Wh
Number of cells	1050	
Weight	73500	g
Commercial Price	4777	EUR

Table 2: HV Battery specs (only considering cells)

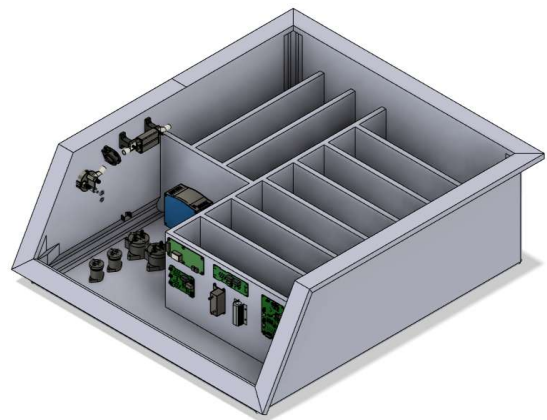
With regard to the battery cooling system, several solutions were evaluated, the most common being forced air cooling and liquid cooling. However, due to competition regulations and safety requirements, specifically the obligation for the assembly to comply with the IP68 protection standard, the water-cooling solution was selected.

To properly size the cooling system, various calculations and simulations were carried out by the responsible department, ensuring sufficient thermal dissipation capacity under both continuous operation and peak power conditions.

In addition, a complete 3D model of the battery was developed (Figure 22), which enabled accurate determination of component dimensions, optimization of mechanical integration, and the design of specific parts. Some of these parts were subsequently manufactured using 3D printing, with the objective of enhancing the structural robustness of the assembly. Another department also did this part.



(a) 3D model of the battery pack



(b) 3D model of the Battery box

Figure 22

5.2.1 Battery Packs Assembly

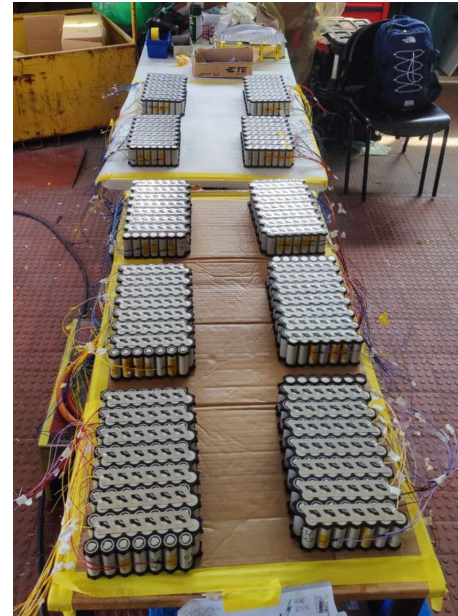
The assembly of the battery began with the measurement of the internal resistance of each individual cell (Figure 23). Based on these measurements, the cells were arranged to ensure that the parallel groups exhibited approximately equal overall resistance. This procedure is essential to guarantee uniform current distribution among parallels, thereby preventing imbalances in discharge that could reduce the effective capacity and overall performance of the battery pack.

After resistance matching, all cells were placed into cell holders, with thermistors positioned along the sides for temperature monitoring. The cells were then interconnected using the spot welder previously described in Section 4.1, specifically the kWeld in combination with the kCap module.

Subsequently, the sensing connections were installed. These consist of nickel tabs placed on top of each parallel group to connect them to the BMS, enabling voltage sensing and cell balancing. Additional thermistors were also mounted on top of the nickel tabs to improve thermal monitoring accuracy.



(a) Measuring internal resistance



(b) Ten packs already soldered

Figure 23

Finally, thermal pads were applied on top of the nickel layer, followed by the installation of cooling plates and 3D-printed structural components designed to secure and reinforce the battery pack assembly.

5.2.2 BMS

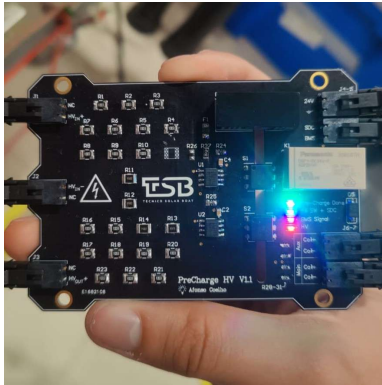
maybe pedir ao Zé Pedro para escrever esta parte

5.2.3 Auxiliary PCB

In addition to the BMS, three custom-designed PCBs play an essential role in the system:

- **Pre-Charge:** Due to the input capacitance of the motor controller, a high inrush current occurs when the power relay closes. To prevent relay welding, which could result in a safety hazard, the pre-charge PCB controls a smaller relay in series with a resistor. This resistor limits the inrush current, allowing the capacitors to charge gradually before the main relay is engaged.
- **Discharge:** The motor controller's input capacitance must also be safely discharged once the system is powered down. This ensures that the high-voltage bus is no longer energized, confining any potential hazard strictly within the battery pack.
- **Tractive System Active Light (TSAL):** This PCB provides an additional safety measure by alerting the driver to the presence of HV outside the battery enclosure. It activates an amber blinking light when high voltage is present, and it can also be configured to display a continuous light when no voltage is detected (this second functionality is currently not in use).

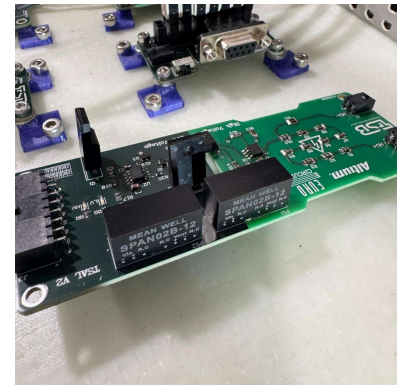
Further details regarding their operation can be found in the dedicated PCB documentation.



(a) Pre-charge



(b) Discharge



(c) TSAL

Figure 24

5.3 Motor

The selected motor was the Beyond Motor AXM2 (Medium Voltage), featuring water cooling and a peak power output of 130 kW. Notably, it is also lightweight, with a mass of only 14.5 kg.

In addition, the motor is equipped with an RLS RM44SI encoder and a KTY 81/210 temperature sensor.

Motor	AXM2	
Winding type	HIGH Voltage (850Vdc)	MEDIUM Voltage (550Vdc)
Peak power	130 kW	
Continuous power	75 kW	
Peak torque	180 Nm	
Continuous torque	120 Nm	
Maximum motor speed	7500 RPM	
Continuous motor speed	6000 RPM	
Peak current	190 Arms	380 Arms
Continuous current	130 Arms	260 Arms
Weight	14,5kg	
Rotor inertia	0,0228 kgm^2	
Kv at no load	8.8	17.6
Kv at continuous torque	7	14
Kv at peak torque	6.2	12.3
Torque constant	0,95 Nm/Arms	0,48Nm/Arms
Motor type	3-phase AC PM	3-phase AC PM

(a) Motor specs



(b) Beyond AXM2

Figure 25

5.4 Motor Controller

As you may know, we can not control the motor by simply powering it with a battery. A motor controller regulates voltage, current, and timing to achieve precise speed, torque, and direction control.

For a three-phase motor, this is even more critical. A simple battery provides only DC, while the motor requires three AC waveforms, phase-shifted by 120° from each other. These must be generated with the correct frequency and amplitude to produce a rotating magnetic field. The controller's role is to convert DC into these synchronized AC signals and manage them dynamically based on load and operating conditions.

So we analysed the market and talked with some companies to understand what would be the most suitable motor controller for us.

These were the characteristics that we were looking for:

- Voltage range within the battery voltage range ($V_{max} = 630V$);
- Compatible with 3 phase PMSM motor;
- Support the peak current of the motor ($A_{peak} = 380Arms$)
- Water cooled;
- Can communication (preferably allowing CAN-id, and bit-rate configuration);
- Compatible with the motor's encoder (RLS RM44SI);
- Compatible with the motor's temperature sensor;

In summary, the motor controller needed to be compatible with the battery and the motor.

In the end, the motor controller chosen was the DTI HV-550, primarily due to its ease of integration with the selected motor.

Low voltage range	9-28	V
High Voltage Range	30-800	V
HV maximum input current Range	280	A
Continuous AC current at 800 Vdc	365	Arms
Maximum AC current for short period (peak current)	550	A_{peak}
Maximum AC current for short period (RMS)	390	Arms
Maximum ERPMS	100000	
Typical efficiency	95	%

Table 3: Motor controller basic specs

The motor controller was installed in the port side of the back compartment in the stringers and is securely fastened to prevent any movement.

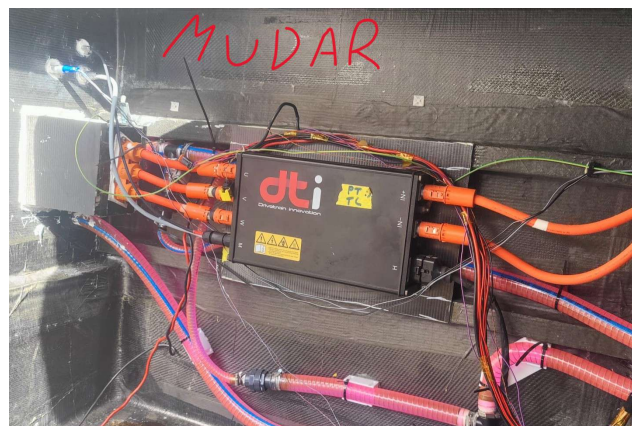


Figure 26: Motor Controller assembly **MUDAR ESTA FOTO!!**

Special care was taken on the connectors assembly to prevent short circuits between phases, shield, ground, or any logical signal. Any short circuit can cause a lot of damage to the components or persons nearby. As for

the encoder and temperature sensor, the shield is essential due to the magnetic interference caused by the AC signals.

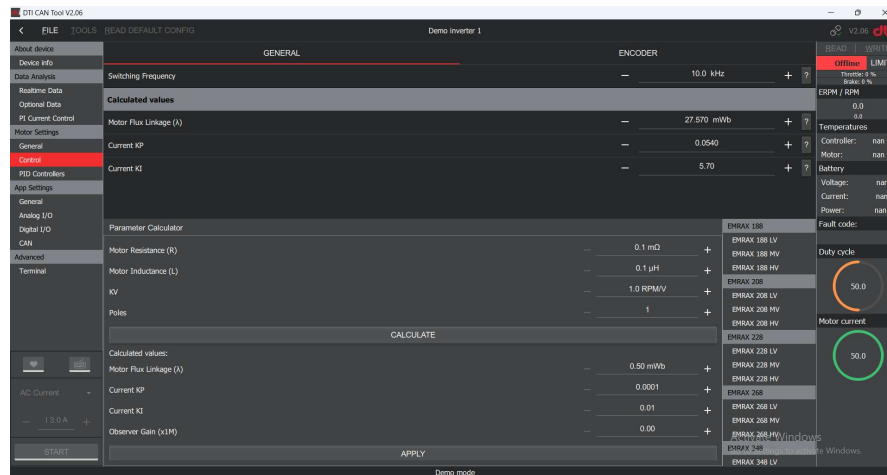


Figure 27: DTI CAN tool GUI

Once the setup is complete, the controller can be configured very easily using the "DTI CAN tool" and its GUI. In this GUI, you can configure the CAN ID, bit rate, encoder settings, PID gains, max and min values of voltage, RPMs, power, temperature, etc.

Some configurations depend on motor parameters such as resistance, inductance, and the number of poles. These values were provided by the motor manufacturer and are not publicly available. In the case of SR03, we estimated these values ourselves in order to optimize the controller parameters. This year, however, we did not have the time or the necessary tools to perform this estimation. Nonetheless, the system is currently operating satisfactorily.

6 Future Work

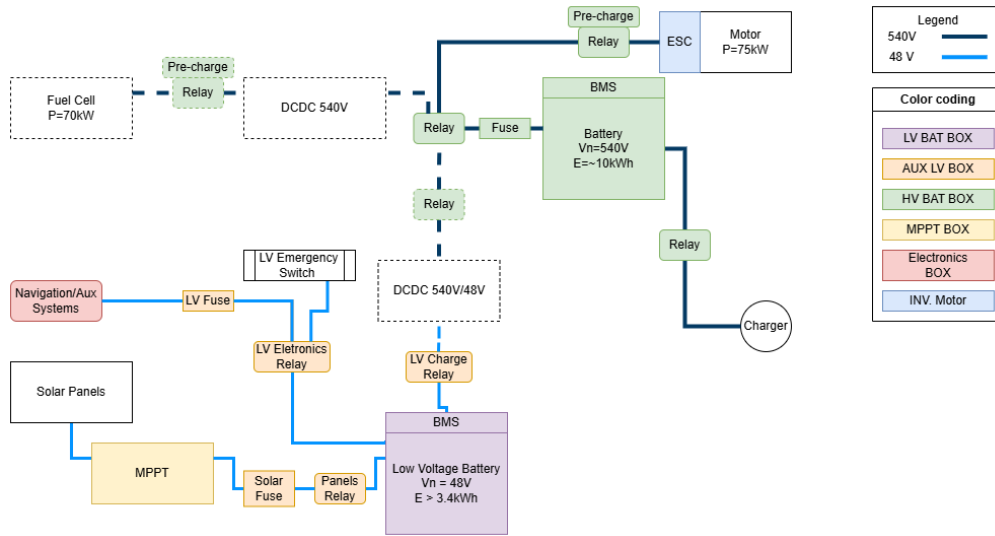


Figure 28: Simplified electrical schematic of SG01 with Fuel Cell

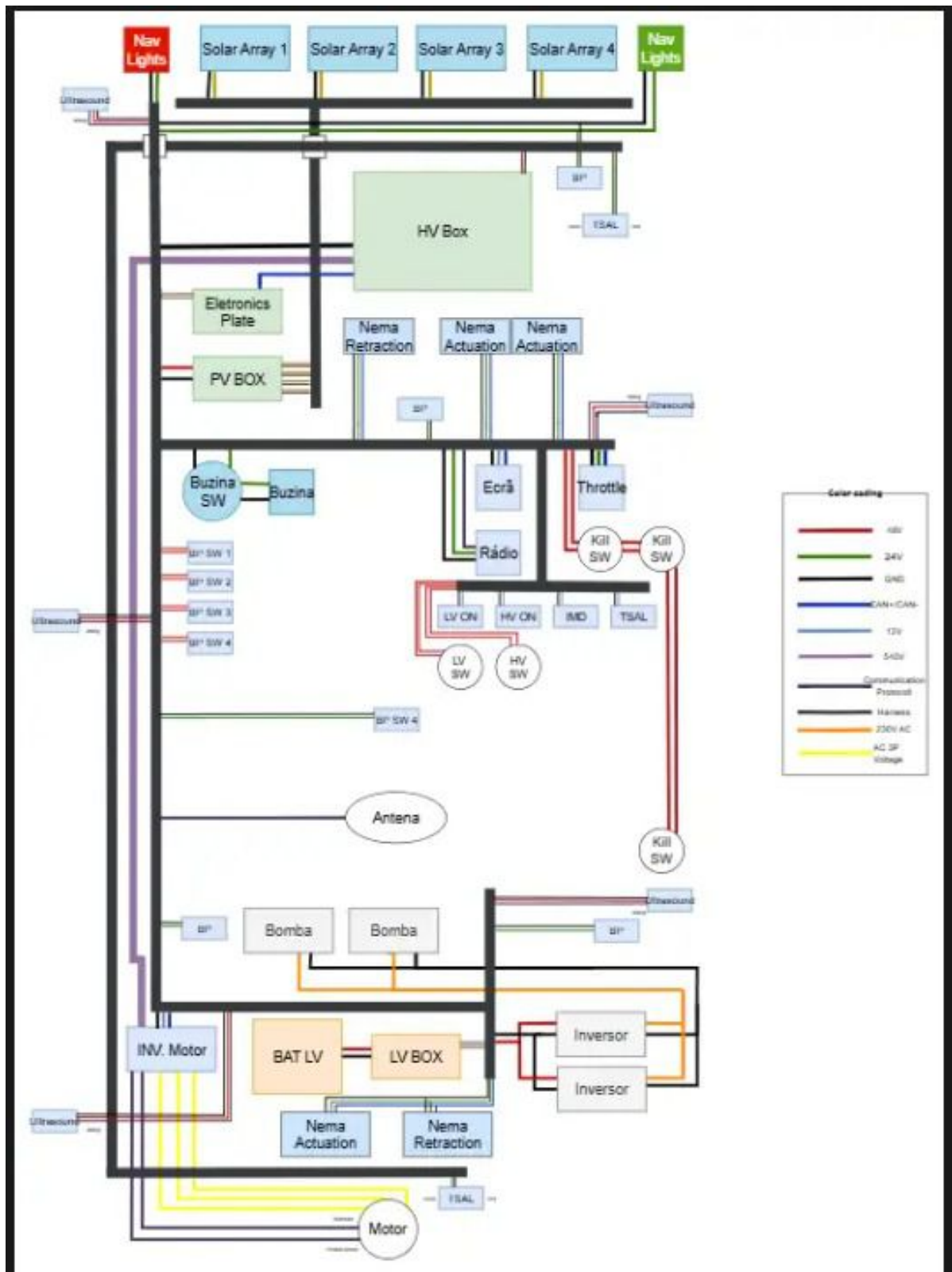
The prototype is not finished, there is still a lot to do to achieve what we planned. We are making some of the missing pieces as you read this document and we will document them in a future revision of this document.

I'll leave some spoilers here:

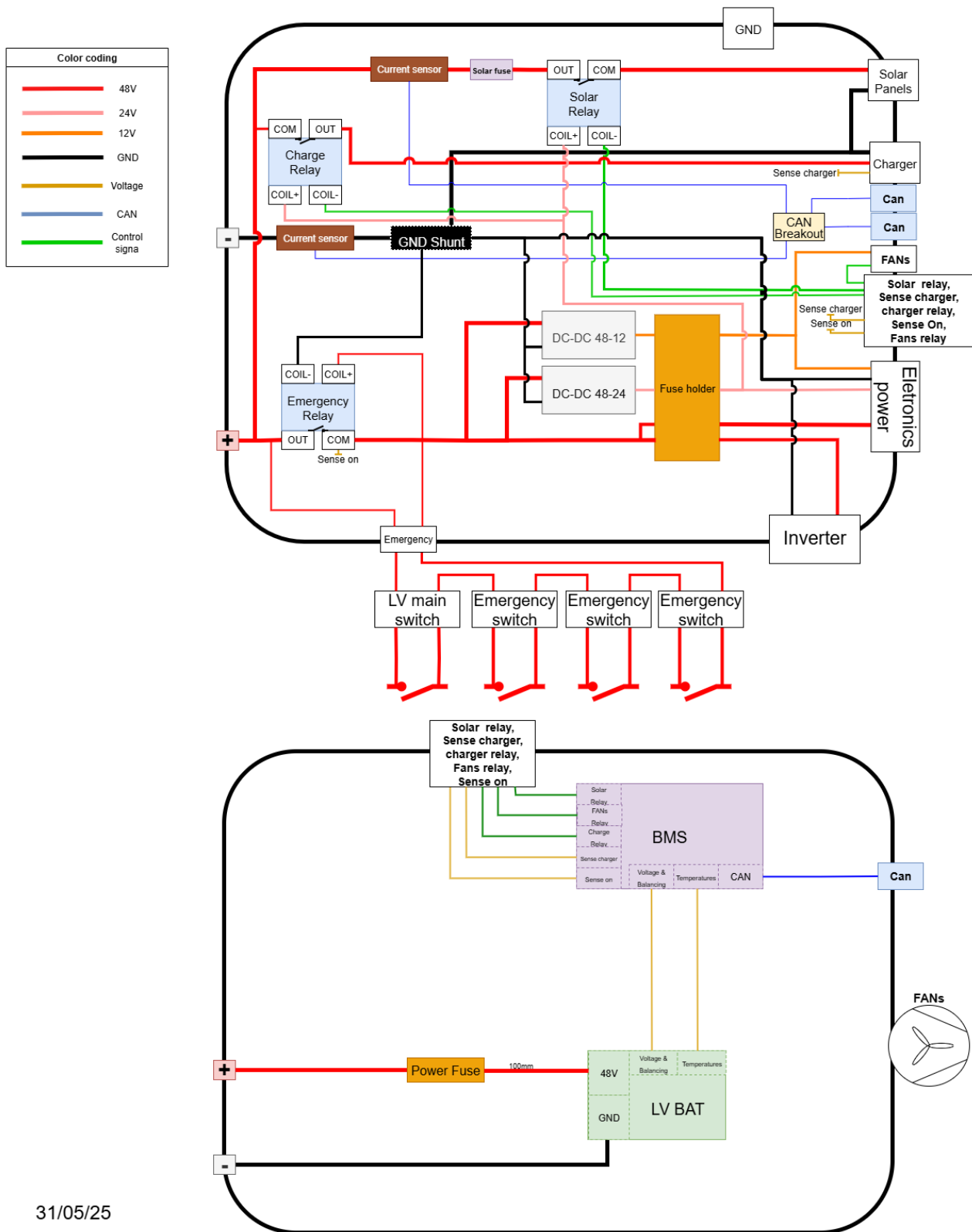
- Fuel Cell integration;
- Custom MPPT with CAN communication;
- Covert some PCBs from previus used MCUs to STM32;
- CSS2 connector to charge the boat in any marina;
- DCDC between HV battery and LV battery;
- New and better Pre-charge;
- Swap the cooling pumps for HV pumps;
- Swap the bilge pumps for 48V or 24V pumps
- Better cooling connectors;
- etc.

We invite you to contribute to the development of these systems by sharing your opinions and suggestions through our social media platforms, website, email, or in person. New members are recruited each semester, so we encourage you to stay connected and engaged.

7 Full Boat Overview

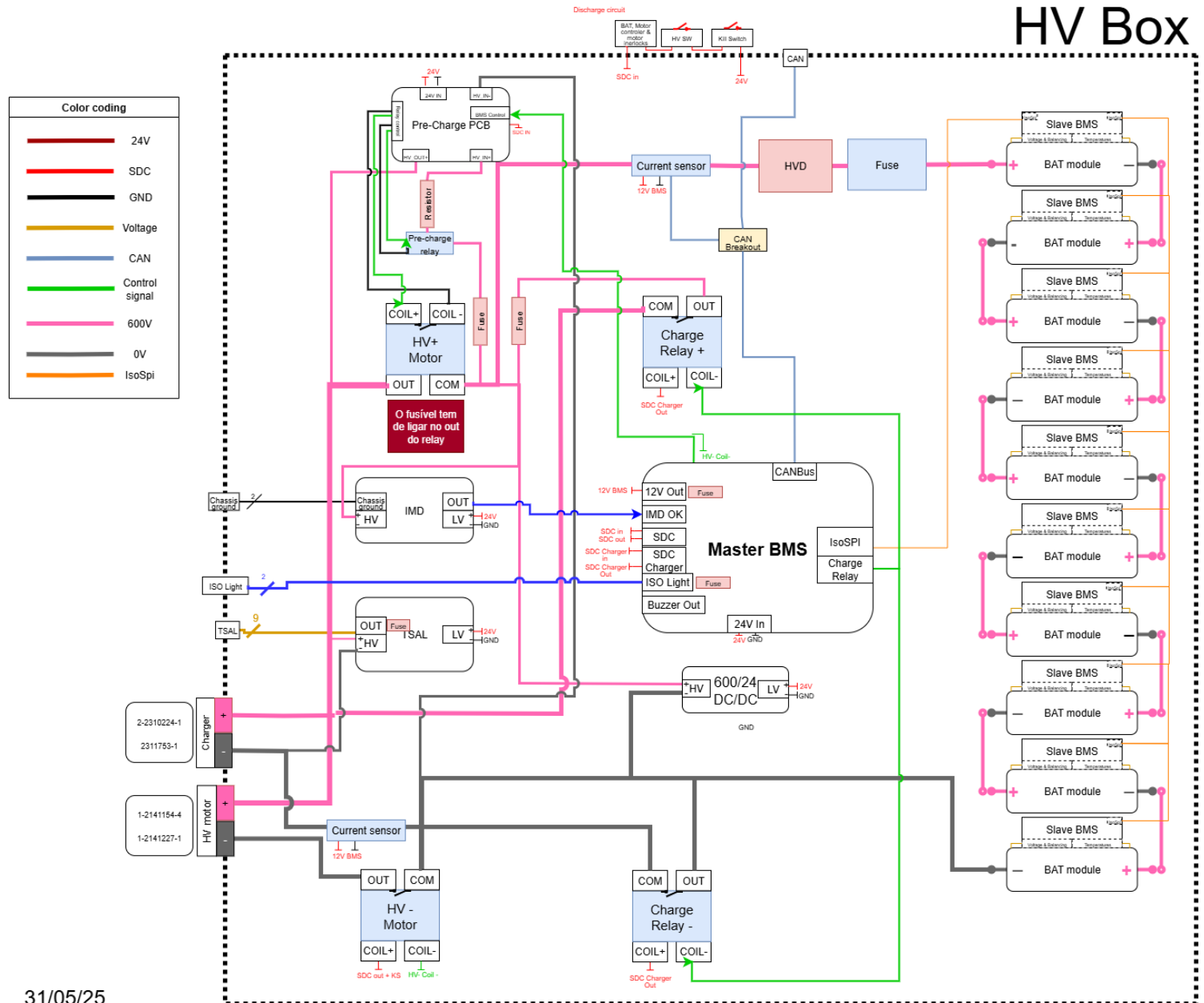


8 LV Battery Box Schematic



31/05/25

9 HV Battery Box Schematic



31/05/25

Electronics plate

